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MSSQL LAB: Telecom Data Mapping

Business Case → Database Fields

Use Case: Telecom Operator Recharge & Data Usage Analytics

=====

1. Create Database

```
8  CREATE DATABASE TelecomDataMappingLab;
9  GO
10
11  USE TelecomDataMappingLab;
12  GO
```

=====

2. Create Tables

=====

Customer master table

```
19  CREATE TABLE customers (
20      customer_id INT IDENTITY(1,1) PRIMARY KEY,
21      customer_full_name VARCHAR(100) NOT NULL,
22      nrc_number VARCHAR(50),
23      gender VARCHAR(10),
24      city VARCHAR(50),
25      created_at DATETIME DEFAULT GETDATE()
26  );
27  GO
```

Subscriber / SIM table

```
30  CREATE TABLE subscribers (
31      subscriber_id INT IDENTITY(1,1) PRIMARY KEY,
32      customer_id INT NOT NULL,
33      msisdn VARCHAR(20) NOT NULL UNIQUE,
34      sim_number VARCHAR(30) NOT NULL,
35      package_type VARCHAR(50),
36      activation_date DATE,
37      status VARCHAR(20),
38      CONSTRAINT fk_subscriber_customer
39          FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
40  );
41  GO
```

Recharge transaction table

```
44 CREATE TABLE recharge_transactions (  
45     recharge_id INT IDENTITY(1,1) PRIMARY KEY,  
46     subscriber_id INT NOT NULL,  
47     recharge_amount DECIMAL(12,2) NOT NULL,  
48     recharge_channel VARCHAR(50),  
49     recharge_created_at DATETIME DEFAULT GETDATE(),  
50     CONSTRAINT fk_recharge_subscriber  
51         FOREIGN KEY (subscriber_id) REFERENCES subscribers(subscriber_id)  
52 );  
53 GO
```

Data usage table

```
56 CREATE TABLE data_usage (  
57     usage_id INT IDENTITY(1,1) PRIMARY KEY,  
58     subscriber_id INT NOT NULL,  
59     usage_date DATE NOT NULL,  
60     total_data_mb DECIMAL(12,2) NOT NULL,  
61     network_type VARCHAR(20),  
62     CONSTRAINT fk_usage_subscriber  
63         FOREIGN KEY (subscriber_id) REFERENCES subscribers(subscriber_id)  
64 );  
65 GO
```

Business-to-database mapping table

```
68 CREATE TABLE data_mapping_catalog (  
69     mapping_id INT IDENTITY(1,1) PRIMARY KEY,  
70     business_term VARCHAR(100) NOT NULL,  
71     source_system VARCHAR(100),  
72     database_table VARCHAR(100),  
73     database_field VARCHAR(100),  
74     field_description VARCHAR(255),  
75     data_type VARCHAR(50),  
76     data_owner VARCHAR(100)  
77 );  
78 GO
```

=====

3. Insert Sample Data

=====

=====

4. Basic SELECT Queries

=====

View all customers

149

SELECT * FROM customers;

150

GO

100 %

No issues found

Results

Messages

	customer_id	customer_full_name	nrc_number	gender	city	created_at
1	1	Aung Aung	12/ABC(N)123456	Male	Yangon	2026-05-11 11:00:47.237
2	2	Su Su	10/XYZ(N)234567	Fem...	Mand...	2026-05-11 11:00:47.237
3	3	Kyaw Kyaw	8/MNO(N)345678	Male	Naypy...	2026-05-11 11:00:47.237
4	4	Hla Hla	7/PQR(N)456789	Fem...	Bago	2026-05-11 11:00:47.237
5	5	Mya Mya	9/STU(N)567890	Fem...	Yangon	2026-05-11 11:00:47.237

View all subscribers

153

SELECT * FROM subscribers;

154

GO

100 %

No issues found

Results

Messages

	subscriber_id	customer_id	msisdn	sim_number	package_type	activation_date	status
1	1	1	09790000001	SIM100001	Prepaid	2025-01-10	Active
2	2	2	09790000002	SIM100002	Postpaid	2025-02-15	Active
3	3	3	09790000003	SIM100003	Prepaid	2025-03-20	Inactive
4	4	4	09790000004	SIM100004	Prepaid	2025-04-05	Active
5	5	5	09790000005	SIM100005	Postpaid	2025-05-12	Active

View all recharge transactions

157

SELECT * FROM recharge_transactions;

158

GO

100 %

No issues found

Results

Messages

	recharge_id	subscriber_id	recharge_amount	recharge_channel	recharge_created_at
1	1	1	5000.00	Mobile Wallet	2026-05-01 09:30:00.000
2	2	1	7000.00	Retail Shop	2026-05-03 14:20:00.000
3	3	2	15000.00	Bank App	2026-05-02 10:10:00.000
4	4	3	3000.00	Retail Shop	2026-05-04 16:45:00.000
5	5	4	12000.00	Mobile Wallet	2026-05-05 11:00:00.000
6	6	5	20000.00	Bank App	2026-05-06 18:30:00.000
7	7	1	10000.00	Mobile Wallet	2026-05-07 08:00:00.000

View all data usage records

161

SELECT * FROM data_usage;

162

GO

100 %

No issues found

Results

Messages

	usage_id	subscriber_id	usage_date	total_data_mb	network_type
1	1	1	2026-05-01	1200.50	4G
2	2	1	2026-05-02	850.25	4G
3	3	2	2026-05-02	2300.75	5G
4	4	3	2026-05-03	400.00	3G
5	5	4	2026-05-04	1750.30	4G
6	6	5	2026-05-05	3200.90	5G
7	7	1	2026-05-06	1500.00	4G

View data mapping catalog

165

SELECT * FROM data_mapping_catalog;

166

GO

100 %

No issues found

Results

Messages

	mapping_id	business_term	source_system	database_table	database_field	field_description	data_type	data_owner
1	1	Customer Name	CRM System	customers	customer_full_name	Full name of telecom customer	VARCHAR(100)	CRM Team
2	2	Customer NRC	CRM System	customers	nrc_number	National registration number	VARCHAR(50)	CRM Team
3	3	Customer City	CRM System	customers	city	Customer registered city	VARCHAR(50)	CRM Team
4	4	Mobile Number	Billing System	subscribers	msisdn	Customer mobile phone number	VARCHAR(20)	Billing Team
5	5	SIM Number	Billing System	subscribers	sim_number	SIM card serial number	VARCHAR(30)	Billing Team
6	6	Package Type	Billing System	subscribers	package_type	Prepaid or postpaid package type	VARCHAR(50)	Product Team
7	7	Recharge Amount	Billing System	recharge_transactions	recharge_amount	Top-up amount by customer	DECIMAL(12,2)	Billing Team
8	8	Recharge Date	Billing System	recharge_transactions	recharge_created_at	Date and time of recharge	DATETIME	Billing Team
9	9	Recharge Channel	Billing System	recharge_transactions	recharge_channel	Channel used for recharge	VARCHAR(50)	Sales Team
10	10	Internet Usage	Network System	data_usage	total_data_mb	Total internet usage in MB	DECIMAL(12,2)	Network Team
11	11	Usage Date	Network System	data_usage	usage_date	Date of internet usage	DATE	Network Team
12	12	Network Type	Network System	data_usage	network_type	3G, 4G, or 5G network type	VARCHAR(20)	Network Team

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5. Business Case 1: Customer and SIM Information

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```

172  SELECT
173      c.customer_full_name AS [Customer Name],
174      c.city AS [City],
175      s.msisdn AS [Mobile Number],
176      s.sim_number AS [SIM Number],
177      s.package_type AS [Package Type],
178      s.status AS [Subscriber Status]
179  FROM customers c
180  INNER JOIN subscribers s
181      ON c.customer_id = s.customer_id;
182  GO

```

100 % No issues found

	Customer Name	City	Mobile Number	SIM Number	Package Type	Subscriber Status
1	Aung Aung	Yangon	09790000001	SIM100001	Prepaid	Active
2	Su Su	Mandalay	09790000002	SIM100002	Postpaid	Active
3	Kyaw Kyaw	Naypyitaw	09790000003	SIM100003	Prepaid	Inactive
4	Hla Hla	Bago	09790000004	SIM100004	Prepaid	Active
5	Mya Mya	Yangon	09790000005	SIM100005	Postpaid	Active

6. Business Case 2: Recharge History

```

188  SELECT
189      c.customer_full_name AS [Customer Name],
190      s.msisdn AS [Mobile Number],
191      r.recharge_amount AS [Recharge Amount],
192      r.recharge_channel AS [Recharge Channel],
193      r.recharge_created_at AS [Recharge Date]
194  FROM customers c
195  INNER JOIN subscribers s
196      ON c.customer_id = s.customer_id
197  INNER JOIN recharge_transactions r
198      ON s.subscriber_id = r.subscriber_id
199  ORDER BY r.recharge_created_at;
200  GO

```

100 % No issues found

	Customer Name	Mobile Number	Recharge Amount	Recharge Channel	Recharge Date
1	Aung Aung	09790000001	5000.00	Mobile Wallet	2026-05-01 09:30:00.000
2	Su Su	09790000002	15000.00	Bank App	2026-05-02 10:10:00.000
3	Aung Aung	09790000001	7000.00	Retail Shop	2026-05-03 14:20:00.000
4	Kyaw Kyaw	09790000003	3000.00	Retail Shop	2026-05-04 16:45:00.000
5	Hla Hla	09790000004	12000.00	Mobile Wallet	2026-05-05 11:00:00.000
6	Mya Mya	09790000005	20000.00	Bank App	2026-05-06 18:30:00.000
7	Aung Aung	09790000001	10000.00	Mobile Wallet	2026-05-07 08:00:00.000

7. Business Case 3: Total Recharge by Customer

```
206  SELECT
207      c.customer_full_name AS [Customer Name],
208      s.msisdn AS [Mobile Number],
209      SUM(r.recharge_amount) AS [Total Recharge Amount]
210  FROM customers c
211  INNER JOIN subscribers s
212      ON c.customer_id = s.customer_id
213  INNER JOIN recharge_transactions r
214      ON s.subscriber_id = r.subscriber_id
215  GROUP BY c.customer_full_name, s.msisdn
216  ORDER BY [Total Recharge Amount] DESC;
217  GO
218
```

100 % No issues found

Results Messages

	Customer Name	Mobile Number	Total Recharge Amount
1	Aung Aung	09790000001	22000.00
2	Mya Mya	09790000005	20000.00
3	Su Su	09790000002	15000.00
4	Hla Hla	09790000004	12000.00
5	Kyaw Kyaw	09790000003	3000.00

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8. Business Case 4: Customers Recharged More Than 10,000 MMK

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```
223  SELECT
224      c.customer_full_name AS [Customer Name],
225      s.msisdn AS [Mobile Number],
226      SUM(r.recharge_amount) AS [Total Recharge Amount]
227  FROM customers c
228  INNER JOIN subscribers s
229      ON c.customer_id = s.customer_id
230  INNER JOIN recharge_transactions r
231      ON s.subscriber_id = r.subscriber_id
232  GROUP BY c.customer_full_name, s.msisdn
233  HAVING SUM(r.recharge_amount) > 10000
234  ORDER BY [Total Recharge Amount] DESC;
235  GO
236
```

100 % ✓ No issues found

Results		Messages	
	Customer Name	Mobile Number	Total Recharge Amount
1	Aung Aung	09790000001	22000.00
2	Mya Mya	09790000005	20000.00
3	Su Su	09790000002	15000.00
4	Hla Hla	09790000004	12000.00

9. Business Case 5: Data Usage by Customer

```
241 SELECT
242     c.customer_full_name AS [Customer Name],
243     s.msisdn AS [Mobile Number],
244     d.usage_date AS [Usage Date],
245     d.total_data_mb AS [Data Usage MB],
246     d.network_type AS [Network Type]
247 FROM customers c
248 INNER JOIN subscribers s
249     ON c.customer_id = s.customer_id
250 INNER JOIN data_usage d
251     ON s.subscriber_id = d.subscriber_id
252 ORDER BY d.usage_date;
253 GO
254
```

100 % No issues found

	Customer Name	Mobile Number	Usage Date	Data Usage MB	Network Type
1	Aung Aung	09790000001	2026-05-01	1200.50	4G
2	Aung Aung	09790000001	2026-05-02	850.25	4G
3	Su Su	09790000002	2026-05-02	2300.75	5G
4	Kyaw Kyaw	09790000003	2026-05-03	400.00	3G
5	Hla Hla	09790000004	2026-05-04	1750.30	4G
6	Mya Mya	09790000005	2026-05-05	3200.90	5G
7	Aung Aung	09790000001	2026-05-06	1500.00	4G

=====

10. Business Case 6: Total Data Usage by Customer

=====

```
259  SELECT
260      c.customer_full_name AS [Customer Name],
261      s.msisdn AS [Mobile Number],
262      SUM(d.total_data_mb) AS [Total Data Usage MB]
263  FROM customers c
264  INNER JOIN subscribers s
265      ON c.customer_id = s.customer_id
266  INNER JOIN data_usage d
267      ON s.subscriber_id = d.subscriber_id
268  GROUP BY c.customer_full_name, s.msisdn
269  ORDER BY [Total Data Usage MB] DESC;
270  GO
```

100 % No issues found

Results Messages

	Customer Name	Mobile Number	Total Data Usage MB
1	Aung Aung	09790000001	3550.75
2	Mya Mya	09790000005	3200.90
3	Su Su	09790000002	2300.75
4	Hla Hla	09790000004	1750.30
5	Kyaw Kyaw	09790000003	400.00

11. Business Case 7: Combined Recharge and Data Usage Summary

```
276  SELECT
277      c.customer_full_name AS [Customer Name],
278      s.msisdn AS [Mobile Number],
279      s.package_type AS [Package Type],
280      SUM(DISTINCT r.recharge_amount) AS [Total Recharge Amount],
281      SUM(d.total_data_mb) AS [Total Data Usage MB]
282  FROM customers c
283  INNER JOIN subscribers s
284      ON c.customer_id = s.customer_id
285  LEFT JOIN recharge_transactions r
286      ON s.subscriber_id = r.subscriber_id
287  LEFT JOIN data_usage d
288      ON s.subscriber_id = d.subscriber_id
289  GROUP BY c.customer_full_name, s.msisdn, s.package_type;
290  GO
```

100 % No issues found

Results Messages

	Customer Name	Mobile Number	Package Type	Total Recharge Amount	Total Data Usage MB
1	Aung Aung	09790000001	Prepaid	22000.00	10652.25
2	Su Su	09790000002	Postpaid	15000.00	2300.75
3	Kyaw Kyaw	09790000003	Prepaid	3000.00	400.00
4	Hla Hla	09790000004	Prepaid	12000.00	1750.30
5	Mya Mya	09790000005	Postpaid	20000.00	3200.90

12. Search Mapping Catalog

Business Question:

Which database field is used for Mobile Number?

```

298 SELECT
299     business_term,
300     source_system,
301     database_table,
302     database_field,
303     field_description,
304     data_type
305 FROM data_mapping_catalog
306 WHERE business_term = 'Mobile Number';
307 GO
308

```

100 % No issues found

	business_term	source_system	database_table	database_field	field_description	data_type
1	Mobile Number	Billing System	subscribers	msisdn	Customer mobile phone number	VARCHAR(20)

13. Search All Billing System Fields

```

313 SELECT
314     business_term,
315     database_table,
316     database_field,
317     field_description
318 FROM data_mapping_catalog
319 WHERE source_system = 'Billing System';
320 GO

```

100 % No issues found

	business_term	database_table	database_field	field_description
1	Mobile Number	subscribers	msisdn	Customer mobile phone number
2	SIM Number	subscribers	sim_number	SIM card serial number
3	Package Type	subscribers	package_type	Prepaid or postpaid package type
4	Recharge Amount	recharge_transactions	recharge_amount	Top-up amount by customer
5	Recharge Date	recharge_transactions	recharge_created_at	Date and time of recharge
6	Recharge Channel	recharge_transactions	recharge_channel	Channel used for recharge

14. Create View for Business Users

```

326 CREATE VIEW vw_customer_recharge_summary AS
327 SELECT
328     c.customer_full_name,
329     c.city,
330     s.msisdn,
331     s.package_type,
332     SUM(r.recharge_amount) AS total_recharge_amount
333 FROM customers c
334 INNER JOIN subscribers s
335     ON c.customer_id = s.customer_id
336 INNER JOIN recharge_transactions r
337     ON s.subscriber_id = r.subscriber_id
338 GROUP BY c.customer_full_name, c.city, s.msisdn, s.package_type;
339 GO

```

Test View

```

342 SELECT * FROM vw_customer_recharge_summary;
343 GO

```

100 % No issues found

	customer_full_name	city	msisdn	package_type	total_recharge_amount
1	Aung Aung	Yangon	09790000001	Prepaid	22000.00
2	Su Su	Mandalay	09790000002	Postpaid	15000.00
3	Kyaw Kyaw	Naypyitaw	09790000003	Prepaid	3000.00
4	Hla Hla	Bago	09790000004	Prepaid	12000.00
5	Mya Mya	Yangon	09790000005	Postpaid	20000.00

15. Stored Procedure: Get Customer Recharge Summary

```

349 CREATE PROCEDURE sp_get_customer_recharge_summary
350 AS
351 BEGIN
352     SELECT
353         customer_full_name,
354         city,
355         msisdn,
356         package_type,
357         total_recharge_amount
358     FROM vw_customer_recharge_summary
359     ORDER BY total_recharge_amount DESC;
360 END;
361 GO

```

Execute Stored Procedure

```
364 EXEC sp_get_customer_recharge_summary;
365 GO
366
```

100 % No issues found

Results Messages

	customer_full_name	city	msisdn	package_type	total_recharge_amount
1	Aung Aung	Yangon	09790000001	Prepaid	22000.00
2	Mya Mya	Yangon	09790000005	Postpaid	20000.00
3	Su Su	Mandalay	09790000002	Postpaid	15000.00
4	Hla Hla	Bago	09790000004	Prepaid	12000.00
5	Kyaw Kyaw	Naypyitaw	09790000003	Prepaid	3000.00

16. Stored Procedure: Search Mapping by Business Term

```
371 CREATE PROCEDURE sp_search_data_mapping
372     @BusinessTerm VARCHAR(100)
373 AS
374 BEGIN
375     SELECT
376         business_term,
377         source_system,
378         database_table,
379         database_field,
380         field_description,
381         data_type,
382         data_owner
383     FROM data_mapping_catalog
384     WHERE business_term LIKE '%' + @BusinessTerm + '%';
385 END;
386 GO
```

Example

```
389 EXEC sp_search_data_mapping 'Recharge';
390 GO
391
```

100 % No issues found

Results Messages

	business_term	source_system	database_table	database_field	field_description	data_type	data_owner
1	Recharge Amount	Billing System	recharge_transactions	recharge_amount	Top-up amount by customer	DECIMAL(12,2)	Billing Team
2	Recharge Date	Billing System	recharge_transactions	recharge_created_at	Date and time of recharge	DATETIME	Billing Team
3	Recharge Channel	Billing System	recharge_transactions	recharge_channel	Channel used for recharge	VARCHAR(50)	Sales Team

17. Data Quality Check

Check missing customer names

```
397  SELECT *
398  FROM customers
399  WHERE customer_full_name IS NULL;
400  GO
```

100 % No issues found

Results Messages

	customer_id	customer_full_name	nrc_number	gender	city	created_at
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Check inactive subscribers

```
403  SELECT *
404  FROM subscribers
405  WHERE status = 'Inactive';
406  GO
```

100 % No issues found

Results Messages

	subscriber_id	customer_id	msisdn	sim_number	package_type	activation_date	status
1	3	3	09790000003	SIM100003	Prepaid	2025-03-20	Inactive

Check recharge amount less than or equal to zero

```
409  SELECT *
410  FROM recharge_transactions
411  WHERE recharge_amount <= 0;
412  GO
```

100 % No issues found

Results Messages

	recharge_id	subscriber_id	recharge_amount	recharge_channel	recharge_created_at
--	-------------	---------------	-----------------	------------------	---------------------

Check data usage less than zero

```
415  SELECT *
416  FROM data_usage
417  WHERE total_data_mb < 0;
418  GO
```

100 % No issues found

Results Messages

	usage_id	subscriber_id	usage_date	total_data_mb	network_type
--	----------	---------------	------------	---------------	--------------

18. Final Business Report

```

424  SELECT
425      c.customer_full_name AS [Customer Name],
426      c.city AS [City],
427      s.msisdn AS [Mobile Number],
428      s.package_type AS [Package Type],
429      s.status AS [Status],
430      ISNULL(SUM(r.recharge_amount), 0) AS [Total Recharge],
431      ISNULL(SUM(d.total_data_mb), 0) AS [Total Data Usage MB]
432  FROM customers c
433  INNER JOIN subscribers s
434      ON c.customer_id = s.customer_id
435  LEFT JOIN recharge_transactions r
436      ON s.subscriber_id = r.subscriber_id
437  LEFT JOIN data_usage d
438      ON s.subscriber_id = d.subscriber_id
439  GROUP BY
440      c.customer_full_name,
441      c.city,
442      s.msisdn,
443      s.package_type,
444      s.status
445  ORDER BY [Total Recharge] DESC;
446  GO

```

	Customer Name	City	Mobile Number	Package Type	Status	Total Recharge	Total Data Usage MB
1	Aung Aung	Yangon	09790000001	Prepaid	Active	66000.00	10652.25
2	Mya Mya	Yangon	09790000005	Postpaid	Active	20000.00	3200.90
3	Su Su	Mand...	09790000002	Postpaid	Active	15000.00	2300.75
4	Hla Hla	Bago	09790000004	Prepaid	Active	12000.00	1750.30
5	Kyaw Kyaw	Naypy...	09790000003	Prepaid	Inact...	3000.00	400.00